

Talha Alam

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Electronics Engineer focused on digital VLSI design and energy-efficient AI hardware acceleration, with hands-on experience both frontend and backend including RTL/FPGA/ASIC development and RTL to GDSII implementation using Cadence EDA industrial tools Genus/Innovus and LibreLane/OpenLane. Experienced in synthesis, floorplanning, placement, clock-tree synthesis, routing, timing analysis, and PPA evaluation. My work spans Gemmini and Systolic Array based Accelerators, CNN hardware, embedded biomedical sensing, IoT systems, and CubeSat-based wireless monitoring. Author of peer-reviewed research and leader of multidisciplinary engineering projects.

EDUCATION

Bachelor of Science in Electronics Engineering | University of Engineering and Technology Peshawar | (2022 – 2026)
Thesis: Extending Gemmini AI Accelerator for Transformer and LLM Acceleration Sep 2025 – July 2026
Designed a Gemmini MOSA with an INT8 Radix-4 Booth compressor multiplier, achieving 20.96/25.19/25.18% area reduction, 2.72 / 20.61 / 38.87% delay improvement for 8×8/16×16/32×32 multipliers, and 674 MHz Fmax. Completed 16×16 Gemmini accelerator RTL to GDSII in Cadence Genus/Innovus, GPDK045 45 nm, with 1.217 mm² footprint.

PUBLICATIONS

- **Talha Alam**, and H. Ali, “A DSP-Free Streaming ASIC Accelerator with Fully Pipelined Dataflow for Real-Time 1D-CNN Inference in Edge-Based Continuous Glucose Monitoring”, **IEEE Transection on Computational System for AI**, 2026, (Under Review)
- **Talha Alam**, and H. Ali, “Hardware-Efficient Hyperglycemic Event Prediction: A Quantized Tiny 1D-CNN for Edge-Based Continuous Glucose Monitoring”, **Advanced Intelligent systems**, Wiley, 2026, (Under Review)
- N. Chalane, M. M. Khan, **Talha Alam**, “Implementation of Stride Prediction and Prefetch Buffering for PicoRV32 RISC-V Processor”, **Proceedings of 6th International Conference on Computational Intelligence & Internet of Things (ICCIoT) 2025**, (Accepted)
- A.A.M.Anik, M.Sadia, A.Tahmid, ..., **Talha Alam**, “Computational Cognitive Network-Based Radiological Mechanism Inspection Technology and Digitally Accessible High-Tech Investigation Robot for Radiological Ground Surveys, Nuclear Environment Inspection, and Radioactive Substance Screening”, **11th International Conference on Advances in Machinery, Material Science and Engineering Application (MMSE 2025)**, 2025, [DOI](#)
- A.A.Mamun, M.Sadia, **Talha Alam**, M.F.Mahmood, “Intelligent Meteorological Observation System to Safeguard Our Ecosystem Amid IR and Piconet Network Protocol”, **11th International Conference on Advances in Machinery, Material Science and Engineering Application (MMSE 2025)**, 2025, [DOI](#)
- **Talha Alam**, M.Javed, A.Rehman, A.A.Mamun, “CanSat: Design, Development, and Deployment of Open-Source Weather Monitoring Nano-Satellite”, **ICoDT 4th IEEE International Conference**, 2024, [DOI](#)

POSTER PRESENTATIONS

- **Talha Alam**, and Hazrat Ali, “Hyperglycaemic Event Prediction through Continuous Glucose Monitoring on Hardware Efficient Deep Learning”, at the 4th UK Multimodal AI workshop, 11-12 June, 2026, London, United Kingdom.

PROFESSIONAL EXPERIENCE

- **Lab Instructor & Teaching Assistant – Digital IC Design & Verification** | *USTP, NSHRDP–INSPIRE (Government of Pakistan)* **July 2026 – Oct 2026**
I have got position of TA in a Government digital design training program for two months to conduct hands-on labs covering the complete RTL to GDSII flow, including RTL design, verification, synthesis, and timing analysis. Guide backend implementation using Cadence, covering floorplanning, placement, STA, routing, physical verification, and GDSII generation.
- **Undergrade Research Assistant** | *University of Stirling | UK (Virtual)* **Oct 2025 – July 2026**
Undergrade research assistant under supervision of Dr Hazrat Ali, focusing on the development of AI for Biomedical on edge FPGA for a low power ASIC application in medical field.
- **Undergrade Research Assistant** | *South Dakota State University | USA (Virtually)* **Dec 2024 – July 2025**
Undergrade research assistant under supervision of Dr Mazhar Sher, focusing on the development of heart failure monitoring devices and smart battery-less systems for agriculture and biomedical applications, integrating IoT and embedded sensing for advanced healthcare and precision farming solutions.
- **Research Intern** | *GIKI | Pakistan* **Jan 2024 – Feb 2026**
Completed research internship at GIKI, focusing on IoT, embedded sensing, and real time data acquisition, contributing to advanced system development and sensor integration.
- **Leadership** | *Aspire Institute | HBS, USA* **Jan 2024 - Apr 2024**

Completed a leadership program at Aspire Institute, developing strategic decision-making, team leadership, and problem-solving skills to drive innovation and impact in technology and research.

- **Research Project Delivery | GPGC | Pakistan**

Nov 2023 - Dec 2023

Designed an IoT-based device for a research project at the Government Post Graduate College's Botany Department, enabling real-time environmental monitoring and data collection for scientific analysis.

RESEARCH PROJECTS

- **AccelClosure – Agentic Prompt to GDSII AI Accelerator Framework (Post)** **September – 2026**
Developed an agentic prompt to silicon framework that automatically generates, verifies, and physically closes configurable AI systolic accelerators from natural language specifications. Evaluated the generated designs using post route PPA, timing/DRC closure, and real Transformer GEMM/attention workloads from TinyBERT, TinyLlama, and Qwen2.5, including real weight and activation INT8/INT32 tensor validation.
- **Optimized Gemmini AI Accelerator – Multiplier to Full SoC (RTL to GDSII) (Post)** **August – 2026**
Designed and implemented a microarchitecture level optimization of the Gemmini AI accelerator using a custom INT8 Radix-4 Booth compressor based multiplier. At 45 nm, the 8×8, 16×16, and 32×32 hierarchical multipliers achieved 20.96%, 25.19%, and 25.18% area reductions with 2.72%, 20.61%, and 38.87% delay improvements, respectively. At accelerator level, 16×16, 32×32, and 64×64 systolic arrays achieved 13.49%, 13.49%, and 12.14% area reductions, with up to 15.49% cell-count reduction and 4.34% delay improvement. Completed Cadence Genus-to-Innovus physical implementation of the proposed 16×16 Gemmini systolic array accelerator, supporting WS and OS dataflows, and generated a 45 nm GDSII with a 1.217 mm² footprint.
- **Mixed Precision Neural Compute Engine (NCE) Accelerator (RTL to GDSII) - (TAPEOUT Project) (Post)** **July – 2026**
Designed, verified, and implemented an 8-lane multi-precision Neural Compute Accelerator using Librelane, yosys, OpenSTA, OpenROAD, KLayout, and SKY130A supporting packed INT8×4, BF16×2, BF24, and FP32 accumulation, with systolic/tiled GEMM, convolution, tensor memories, and AXI4-Lite control; passed 62/62 verification suites and 1,006 AXI transactions. Completed the full SKY130 RTL-to-GDSII flow, delivering an 8.2 × 8.2 mm chip with 154 GPIOs, 16 power/ground pads, 5,999 vias, and zero detailed-routing violations.
- **INT4 Transformer Attention Systolic Array Accelerator (Post)** **May – 2026**
Designed and verified a 16×16 INT4 Transformer attention-tile accelerator with fixed-point softmax and 8-bit memory-mapped wrapper; validated on real GPT-2/distilgpt2 and TinyBert attention tiles, achieving 913 core cycles and 23.18× full-wrapper cycle speedup vs RISC-V.
- **CNN Systolic Array Accelerator ASIC (RTL to GDSII) (Post)** **April – 2026**
Systolic Array-based 1D-CNN accelerator from RTL to GDSII using Cadence Genus/Innovus in GPD045 45nm. Closed post-route timing at 166.7 MHz with 0.164 mm² core area, 64.227% density, and zero setup/hold, transition, capacitance, and antenna violations.
- **High Performance Tiny 1D CNN Accelerator ASIC for CGM Applications (RTL to GDSII) (Post)** **Mar – 2026**
Verilog, SkyWater 130nm PDK, Librelane, Yosys, OpenROAD, KLayout, Area: 21.79 mm² (core), Total Power: 0.216 mW, Timing Closure: 0 setup/hold violations @ 35ns clock
- **INT8 1D CNN Accelerator IP Design and Implementation on Arty S7-35T FPGA (Post)** **Feb – 2026**
Designed an end-to-end INT8 AI accelerator IP by training the model, quantizing it to INT8, and implementing the custom hardware accelerator on the Arty S7-35T FPGA. Post-implementation results: 3,953 LUTs, 4,506 FFs, 42 DSPs, 33 BRAMs, timing met, with reported on-chip power of approximately 3 mW.
- **Streaming Line-Buffer and Dataflow Optimized Tiny 1D-CNN Accelerator for Real-Time Glucose Prediction** **Jan - 2026**
FPGA (Artix-7), Vitis HLS, Vivado, Micro-Architectural Optimizations: Zero DSP Usage, Loop Unrolling, Pipelining, Sliding Window Buffer, 3x Latency Improvement, 21% Power Reduction
- **INT8 Quantization and Optimization of a Tiny 1D-CNN for Glucose (T1 - CGM) Prediction** **Dec - 2025**
INT8 Static Quantization, PyTorch, C/RTL Co-simulation, 1D-CNN, DiaData Dataset (6.2M samples) - [Dataset](#)

SKILLS

Technical Skills: VLSI, Hardware accelerator (ML/DL), ASIC, Cadence Genus/Innovus, 45 nm PDK, 55 nm PDK, 130 nm PDK, Librelane, Floorplanning, RTL to GDSII, Bash, git, microcontrollers, PCB design, and Embedded programming. Experienced in embedded systems, sensor integration, and wireless communication (LoRa, BLE). Skilled in Python, AI, ML on edge devices, C/C++, and MATLAB for automation and data analysis.

Soft Skills: Strong abilities in problem-solving, project management, teamwork, leadership, innovation, and critical thinking. Experienced in technical writing, research, and hands-on prototyping for IoT and biomedical applications.

REFERENCES

Dr. Anees Ullah Assistant Professor, Department of Electronics Engineering University of Engineering and Technology Peshawar, Pakistan Email: aneesullah@uetpeshawar.edu.pk	Dr. Hazrat Ali Assistant Professor, Computing Science University of Stirling, UK Email: ali.hazrat@stir.ac.uk	Dr. Ali Zahir Associate Professor Department of Computer Engineering Comsats University Islamabad, Abbottabad Campus. Email: alizahir@cuiatd.edu.pk	Dr. Mazhar Sher Assistant Professor, Department of Agriculture and biosystem Engineering South Dakota State University, USA Email: msher2015@fau.edu
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